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Risk Evaluation and Management of CO_2 Storage in the Bahrain Oil Field

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Abstract

This paper presents a risk-focused assessment of CO_2 storage in the Mauddud reservoir of the Bahrain Oil Field, emphasizing subsurface uncertainty and technical risk management. A detailed geological model was constructed using core analysis, well log interpretation, and formation micro-imaging (FMI), with the secondary gas cap (248,000 acre-ft) identified as the primary containment unit for CO_2 injection.

A volumetric-based storage estimation was coupled with a Monte Carlo simulation using Oracle Crystal Ball, resulting in a mean CO_2 storage potential of 37.57 million metric tons (MMt), with P10 = 26.88 MMt and P90 = 49.47 MMt. A sensitivity analysis, conducted using both rank correlation and a machine learning-based Random Forest model, revealed that storage efficiency, porosity, and formation volume factor were the dominant parameters influencing storage capacity, while irreducible water saturation exhibited minimal impact.

A Bowtie risk assessment framework was used to identify and evaluate technical risks, including caprock integrity under pressure buildup, fault reactivation near injection zones, potential wellbore leakage due to aging completions, and uncontrolled plume migration. Targeted mitigation strategies such as zonal isolation, corrosion-resistant completions, and real-time pressure monitoring were proposed to address these risks.

This integrated approach, combining probabilistic modeling, machine learning insights, and a structured risk evaluation framework, enhances confidence in safe, long-term CO_2 storage and offers a replicable model for future carbon capture and storage (CCS) projects in similar carbonate reservoirs.

Keywords: CO2 Storage, Risk Evaluation, Machine Learning, Monte-Carlo Simulation, CO2 Storage Efficiency

Introduction

As the world takes urgent steps to address the climate crisis, carbon capture and storage (CCS) has become one of the leading strategies in the global push to lower emissions. While renewable energy and efficiency improvements are long-term solutions, CCS offers a more immediate way to deal with large-scale carbon dioxide (CO_2) emissions especially from industrial sources that are harder to decarbonize. One of the most promising places to store this captured CO_2 is in geological formations deep underground (Bachu, 2003; Wei et al., 2023).

Among the different storage options, depleted oil and gas reservoirs stand out. These reservoirs have already proven they can hold hydrocarbons for millions of years, and thanks to decades of oilfield development, we often have a wealth of data on their structure, sealing integrity, and fluid flow behaviour. Many also have existing infrastructure wells, seismic data, core samples which can significantly reduce the cost and complexity of CCS projects (Iskandar and Usman, 2011; Wei et al., 2023).

Globally, there have already been some success stories. Projects like Weyburn in Canada, In Salah in Algeria, and Cranfield in the U.S. have safely stored millions of tons of CO_2, while also teaching the industry valuable lessons about storage behaviour, monitoring technologies, and risk management (Preston et al., 2005; Freifeld et al., 2013). These projects show that with the right planning and geological understanding, CO_2 storage in depleted reservoirs is not only feasible it's reliable and effective.

In the Middle East, CCS is starting to gain attention as a strategic opportunity. The region's energy systems are deeply rooted in oil and gas production, but the geological conditions and infrastructure make it ideally suited for storing CO_2 as well. This is especially true for fields like the Bahrain Field also known as Awali which was the first oil discovery in the Gulf region. Production began in 1932, and over the decades, the field has seen a wide range of recovery techniques, including waterflooding and extensive gas injection programs that began as early as 1938 (Nemmawi et al., 2018).

The Mauddud Formation within the Bahrain Field is the focus of this study. This carbonate reservoir has a long production history and a complex geological makeup. Over time, pressure changes, aquifer interactions, and fault movements have reshaped how fluids behave in the reservoir. These dynamic changes provide a rare and valuable opportunity: by looking at how gas has moved through the Mauddud over the past 80 years, we can make more accurate predictions about how injected CO_2 might behave in the same environment (Nemmawi et al., 2018).

A key part of this study is estimating how much CO_2 the reservoir can actually hold. To do this, we combine volumetric methods with probabilistic modelling, using tools like Monte Carlo simulations to deal with the natural uncertainty in subsurface data. Parameters like porosity, reservoir thickness, and residual oil saturation are treated as ranges rather than fixed values, which allows us to generate P10, P50, and P90 storage scenarios giving a full picture of best-case, most-likely, and worst-case outcomes (Iskandar and Usman, 2011; Wei et al., 2023).

Beyond just estimating capacity, the study also takes a deep dive into risk evaluation. Using a Bowtie analysis framework, we assess the likelihood and impact of key risks like caprock failure, fault reactivation, and leakage through legacy wellbores. These are serious concerns, and they've been identified in other international projects as well. But the Bahrain Field offers some advantages, especially its thick, laterally extensive caprock and decades of field monitoring and production data (Wei et al., 2023; Nemmawi et al., 2018).

What makes this case study especially compelling is how it combines old and new: a mature field with a rich data history, analyzed using modern tools and techniques. By linking static geological models with historical gas behaviour and probabilistic simulations, this research doesn't just show that CO_2 storage is possible in the Mauddud it provides a roadmap for how to do it safely and effectively.

Ultimately, this work contributes to the broader effort of making CCS a practical and scalable solution in the Middle East. It offers a model for how other mature reservoirs in the region and globally can be reassessed not just as depleted oil fields, but as future CO_2 storage assets.

Overview of the Bahrain Field

The Bahrain Field, more commonly known as the Awali Field, holds a special place in the history of petroleum development in the Gulf. Discovered in 1932, it was the first commercial oil field in the region

—long before discoveries in neighbouring countries reshaped the global energy landscape. What began as a bold exploration effort in the early 20th century has evolved into a mature field with nearly a century of production, reservoir management, and operational experience (Al-Muftah et al., 2003).

Geologically, the field is situated on a doubly plunging anticline that runs through the central and southern parts of Bahrain Island. This structure, along with a sequence of Cretaceous carbonates, has allowed the accumulation and trapping of hydrocarbons in multiple stacked reservoirs. Among these, the Mauddud Formation has proven to be one of the most prolific and technically intriguing. Over the years, Awali has produced from 15 separate reservoirs, each with its own complexities and behaviours but Mauddud remains central due to its high porosity, layered structure, and dynamic pressure history (Nemmawi et al., 2018).

The Mauddud reservoir itself is made up of two primary intervals: the Mauddud A (upper) and Mauddud B (lower) units. These are separated by a thin, often low-resistivity carbonate interval that can vary in porosity and permeability. Over time, leaching and diagenetic processes have created vuggy and channel-like porosity, contributing to the reservoir's high permeability zones—often referred to locally as "fast zones" or "thief zones." These features have long influenced fluid flow behaviour in the field and continue to pose challenges and opportunities for injection and storage strategies (Tatweer Petroleum, 2018).

From a development standpoint, Awali has seen nearly every major production technique applied. It began with natural depletion in the 1930s, followed by gas injection starting as early as 1938 to maintain reservoir pressure. Rich gas was initially sourced from the Arab Formation, and later from the Khuff, to support pressure maintenance in the Mauddud and other units. These gas injection programs were not only among the first in the region, but they also laid the groundwork for how pressure support and fluid migration are understood today in the Bahrain Field.

The long history of injection has significantly changed the fluid distribution in Mauddud. Over the decades, gas has migrated through high-permeability pathways, forming a secondary gas cap in many regions. At the same time, structural adjustments and possible seal weakening have led to lateral and vertical fluid movement, with hydrocarbons migrating into adjacent formations such as the Nahr Umr and Wara. These movements have caused complexities in defining fluid contacts and estimating original oil in place, especially with shifting interpretations of oil-water contacts and residual saturation levels (Nemmawi et al., 2018).

The drive mechanisms in the field have also evolved. Initially dominated by solution gas drive and gravity drainage, they have since been influenced by a combination of gas injection, aquifer support, and fault-assisted migration. This dynamic interaction has made it essential to use detailed geologic and simulation models to predict performance and inform field development strategies. The legacy of high-permeability zones combined with structural compartmentalization has made plume migration modelling for CO_2 storage especially relevant here.

Today, the Bahrain Field presents an almost ideal setting for evaluating the feasibility of CO_2 storage. Its known structure, well-documented production history, and large dataset of core, log, and seismic data provide a strong foundation for modelling and simulation. The Mauddud Formation, in particular, offers both technical challenges and advantages: while heterogeneity and legacy gas migration introduce complexity, the reservoir's thick caprock, high porosity, and existing infrastructure make it a strong candidate for permanent CO_2 sequestration.

This combination of legacy and potential makes the Bahrain Field not just a case study but a regional blueprint for how other mature fields in the Gulf might evolve from oil producers to carbon sinks, playing a new role in the global energy transition.

Key Highlights of the Bahrain Field

1. **First oil discovery in the Gulf.** Discovered in 1932 and brought onstream in 1933, the Bahrain Field established the region's early petroleum industry.

2. **Multi-reservoir system.** Comprises 15 vertically stacked carbonate and clastic reservoirs. The Mauddud Formation stands out as one of the most prolific and technically complex.
3. **Doubly plunging anticline structure.** The field's structural geometry provides a robust trap for hydrocarbons and enhances long-term containment for CO_2 storage.
4. **Long-standing gas injection history.** Rich gas injection began in 1938 and expanded in the 1970s using dry gas from the Khuff Formation offering over 80 years of real-world injection data.
5. **Complex fluid migration behaviour.** Vertical and lateral movement of fluids triggered by pressure changes, aquifer support, and fault activity offers a natural analogue for CO_2 plume dynamics.
6. **Highly heterogeneous reservoir characteristics.** The Mauddud exhibits vuggy porosity, channelized intervals, and low resistivity "fast zones" acting as thief layers, significantly influencing injection strategies.
7. **Regionally extensive caprock systems.** The overlying Wara Formation and underlying Nahr Umr Shale provide thick, laterally continuous seals critical for CO_2 containment and risk minimization.
8. **Rich historical and geological data availability.** Includes decades of well logs, seismic surveys, core studies, production data, and simulation models enabling high-resolution reservoir characterization.
9. **Established infrastructure.** Surface and subsurface facilities (e.g., legacy wells, injection pipelines) reduce costs and accelerate readiness for CCS deployment.
10. **Regional strategic relevance.** Serves as a model for transitioning mature Middle Eastern fields into low-carbon assets through CO_2 sequestration and emissions reduction.

Geological Setting

The Mauddud Formation, located within the prolific Bahrain (Awali) Field, represents a shallow to open marine carbonate reservoir of mid-Cretaceous (Albian) age. It forms part of the larger Wasia Group and is bracketed stratigraphically by the underlying Nahr Umr Shale and the overlying Wara Formation—both of which play critical roles in providing vertical sealing capacity for CO_2 storage.

Structurally, the Mauddud lies within a doubly plunging anticline, shaped by long-term tectonic activity and episodic fault reactivation. The field's crest and flanks exhibit distinct variations in thickness, with the Mauddud averaging around 110 feet but ranging from ~100 ft at the crest to ~122 ft on the flanks (Tatweer Petroleum, 2018). Over time, this structure has undergone subtle deformation and vertical fluid redistribution, making it a geologically dynamic environment for studying injection behaviour (Nemmawi et al., 2018).

The reservoir is typically subdivided into two main intervals: Mauddud A (upper) and Mauddud B (lower). Mauddud A is generally more porous and permeable, with an average thickness of ~75 ft, while Mauddud B tends to be tighter and thinner (~35 ft). These two zones are often separated by a transitional carbonate interval exhibiting lower resistivity and variable porosity, which sometimes acts as a localized flow barrier or thief zone depending on diagenetic overprinting.

From a petrological perspective, the Mauddud is composed predominantly of bioclastic limestone, characterized by grain-dominated packstones, wackestones, and localized rudist-rich debris flows. Over time, the reservoir rock has been altered through extensive diagenesis, including leaching, calcite cementation, recrystallization, and dolomitization. These processes have contributed to the development of vugular and moldic porosity, often forming high-permeability channels that create complex flow pathways (Nemmawi et al., 2018).

One of the defining features of the Mauddud Formation is the presence of "fast zones" intervals where high permeability is concentrated due to vuggy and channel-like porosity. These zones often exhibit low resistivity in log responses, misleading traditional interpretations of saturation and pay. Recent work, including core studies and formation microimager (FMI) analysis, suggests that these fast zones may play a dominant role in controlling both gas and CO_2 movement through the reservoir.

Caprock integrity in this setting is primarily provided by the Wara Formation, a thick sequence of tight shales and silty carbonates, and the underlying Nahr Umr, a known regional seal. Both formations exhibit low permeability and have demonstrated the ability to retain hydrocarbons and injected gas over decades of pressure cycling making them strong candidates to confine supercritical CO_2 over geological timescales.

Faults are present within the Mauddud system, particularly near the flanks of the anticline. These faults appear to have controlled some of the historical fluid migration and may influence plume distribution during injection. However, most faults are thought to be sealing or partially sealing due to mineralization and low displacement, although detailed risk assessments are required near structurally complex zones (Nemmawi et al., 2018).

Overall, the Mauddud reservoir combines favourable storage capacity with a robust containment system, though it presents modelling challenges due to heterogeneity, dual porosity systems, and complex wettability behaviour. Its long production and injection history offer a valuable analogue for predicting CO_2 movement, pressure evolution, and long-term storage performance.

Rock and Fluid Properties of the Mauddud Reservoir (Bahrain Field)

In the absence of explicit property datasets within the provided documentation, the rock and fluid characteristics of the Mauddud reservoir in the Bahrain Field were compiled based on a synthesis of published literature, including peer-reviewed SPE publications and regional geological surveys. The following table presents a consolidated set of reservoir parameters, derived from documented field measurements, core analyses, and analogue carbonate systems within the Arabian Gulf region. These values reflect both historically observed conditions and established ranges typically encountered in mid-Cretaceous carbonate reservoirs of similar depositional and diagenetic environments, Table 1 is presenting the most important data for the Mauddud reservoir.

Table 1—Rock and Fluid Properties of the Mauddud Reservoir in Bahrain Field.

Property	Typical Range / Value	Notes
Depth (TVD)	5,800 – 6,300 ft	Varies crest to flank (Al-Muftah et al., 2003)
Lithology	Bioclastic limestone, rudist-bearing carbonate	Grain-supported, with packstones/wackestones
Porosity (core/log)	15% – 31%	Higher near crest; vuggy and moldic porosity zones common (Nemmawi et al., 2018)
Permeability	10 – 500 mD	"Fast zones" may exceed 1 Darcy locally
Reservoir Thickness	100 – 122 ft (net); (110 ft avg.)	Divided into Mauddud A and B (Tatweer Petroleum, 2018)
Water Saturation (Sw)	10% – 15%	Variable; impacted by residual oil and gas cap effects
Residual Oil Saturation (Sor)	20% – 30%	Core-derived; used in CO_2-EOR/CCS calculations
Gas Saturation (Sg)	Up to 35% in gas cap zones	From legacy gas injection behaviour
Initial Reservoir Pressure	2,400 – 2,700 psi	Depleted due to long production history
Bubble Point Pressure (Pb)	1,900 – 2,100 psi	Indicates slightly undersaturated reservoir initially
Formation Temperature	130°F – 150°F (54°C – 66°C)	Mesothermal
Oil Gravity (API)	24° – 31° API	Medium gravity (Iskandar & Usman, 2011)
Oil Viscosity	1.5 – 2.2 cP	Lower near crest
Formation Water Salinity	100,000 – 130,000 ppm TDS	High-salinity brine typical of Mesozoic Gulf carbonates
Caprock Integrity	High; Wara & Nahr Umr formations act as seals	Confirmed by injection containment history
Wettability	Mixed-wet to weakly oil-wet	Alters CO_2/oil and CO_2/water displacement efficiency
Compressibility (rock)	5 – 15 1/psi	Estimated from carbonate analogues

Methodology

The methodology adopted in this study combines geological characterization, volumetric analysis, and uncertainty quantification to assess the CO_2 storage potential of the Mauddud reservoir. The workflow begins with geological modelling based on core analysis, well log interpretation, and Formation MicroImager (FMI) data to define reservoir architecture and heterogeneity. This is followed by volumetric estimation of Original Oil in Place (OOIP), Gas in Place (GIP), and CO_2 storage capacity using deterministic methods. To address parameter uncertainties, a Monte Carlo simulation was performed using Crystal Ball™ software, generating probabilistic distributions for storage capacity. Finally, a sensitivity analysis was conducted on key reservoir parameters, including area, thickness, porosity, initial water saturation (Swi), and residual oil saturation (Sor) to identify the most influential factors affecting storage estimates and support robust decision-making under uncertainty.

Geological modelling from core, well logs, and FMI imaging

To accurately represent the heterogeneity and structural complexity of the Mauddud reservoir, a multi-disciplinary geological modeling workflow was employed. This process integrates data from core analysis, well log interpretation, and Formation MicroImager (FMI) imaging, each contributing critical information for building a robust 3D static model suitable for CO_2 storage assessment, see Figure 1. The key steps in this workflow are summarized below:

- **Core Analysis**
  - Lithofacies identified from core descriptions to characterize depositional environments.
  - Porosity, permeability, and pore type data derived from laboratory core measurements.
  - Provides ground-truth data for calibrating log interpretations.
- **Well Log Interpretation**
  - Utilizes gamma ray, neutron-density, and resistivity logs.
  - Dual-porosity trends evaluated to differentiate matrix and vuggy contributions.
  - Electrofacies classification performed for facies-based property modelling.
- **FMI (Formation MicroImager) Data Analysis**
  - Used to detect fractures, bedding planes, and high-resolution sedimentary structures.
  - Dip and azimuth analysis help define stratigraphic and structural orientation.
  - Channel and flow feature mapping supports understanding of preferential pathways.
- **Integration for Reservoir Architecture**
  - Combined datasets are used to define internal architecture and heterogeneities.
  - Structural and stratigraphic elements aligned to build realistic reservoir frameworks.
- **3D Geological Model Construction**
  - A geocellular static model is developed incorporating lithofacies, petrophysics, and structures.
  - Model resolution matched to data availability and heterogeneity scale.
- **Simulation Readiness**

- Final model provides the spatial basis for dynamic simulation and CO_2 storage evaluation.
- Ensures representation of key features such as high-permeability zones and caprock continuity.

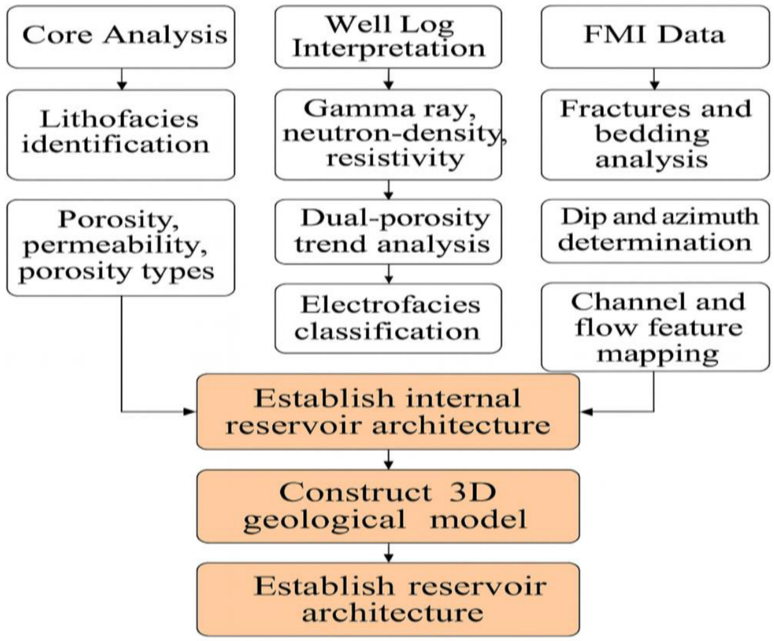

Figure 1—The geological model for CO_2 storage complexity in the Mauddud reservoir

Estimating CO_2 Storage Capacity from Secondary Gas Cap Volume

The estimation of CO_2 storage capacity in depleted or partially depleted gas caps provides a practical framework for evaluating the feasibility of geological carbon storage in mature carbonate reservoirs. In this study, the current volume of the secondary gas cap—formed as a result of gas injection—was used as a proxy to evaluate the available pore space for CO_2 storage in the Mauddud reservoir. This approach assumes that the gas-invaded volume, having already demonstrated containment and injectivity, represents a realistic and technically accessible storage domain.

To ensure alignment with international best practices, the volumetric estimation method adopted is based on the guidelines set forth by the IEA (2020) and Anon (1981), which incorporate key geological and reservoir engineering parameters, including porosity, irreducible water saturation, formation volume factor, CO_2 density under reservoir conditions, and a conservative storage efficiency factor. This method allows for a physically grounded, site-specific quantification of CO_2 storage potential, supporting strategic planning for carbon capture and storage (CCS) in the region.

Injected Gas Volume in the Secondary Gas Cap. The **pore volume** of the current gas cap (after shrinkage) was calculated using:

$$PV = \text{ Gas Cap Volume } \times \phi * PV$$

Given:

- Gas Cap Volume = 248,000 acre-ft
- Porosity $\phi = 0.31$
- Conversion: 1 acre-ft = 43,560 ft³

- Formation Volume Factor for gas Bg = 0.0048 res ft³/scf

PV = 248,000 × 0.31 = 76,880 acre-ft
PV = 76,880 × 43,560 = 3,348,172,800 ft³
Gas Injected = PV/Bg = 3.35×10E9/0.0048 = 697.7 Bscf

CO_2 Storage Capacity in the Gas Cap Volume. To estimate the mass of CO_2 that can be stored in the current gas cap volume (248,000 acre-ft), the following equation was applied, derived from IEA (2020) and Anon (1981):

$M\ CO_2 = V \times \phi \times (1-Swi) \times B \times \rho\ CO_2 \times E$

Step-by-Step Calculation

Step 1: Convert gas cap volume from acre-ft to cubic meters (m³):

V = 248,000 acre-ft × 1233.48 (m³/acre-ft) = 305,903,040 m³

Step 2: Apply porosity and irreducible water saturation:

Effective Pore Volume = 305,903,040 × 0.31 × (1 - 0.11) = 84,309,624 m³

Step 3: Apply formation volume factor (B) = 1.5, CO_2 density (ρ) = 600 kg/m³, and storage efficiency E = 0.5:

$M\ CO_2$ = 84,309,624 × 1.5 × 600 × 0.5 = 37,979,391,931 kg

Convert the result to Million Metric Tons (MMt):

M_CO_2 = 37,979,391,931 ÷ 1,000,000 = 37.98 MMt

Machine Learning-Based Sensitivity Analysis Approach

To complement traditional statistical techniques, a machine learning-based sensitivity analysis was performed using the Random Forest Regressor algorithm. This ensemble learning method is well-suited for capturing complex, non-linear interactions between input parameters and the output variable in this case, the estimated CO_2 storage capacity. By training the model on a synthetic dataset generated through Monte Carlo sampling, the algorithm evaluates the relative importance of each parameter in driving output variability. The computed feature importance scores reveal which reservoir and volumetric factors (such as porosity, formation volume factor, storage efficiency, and irreducible water saturation) most significantly impact the predicted storage potential. This technique enhances interpretability and supports data-driven prioritization of uncertainty in subsurface storage assessments. This machine learning approach highlights the **relative importance** of each parameter used in the volumetric equation and showed the results in Figure 2.

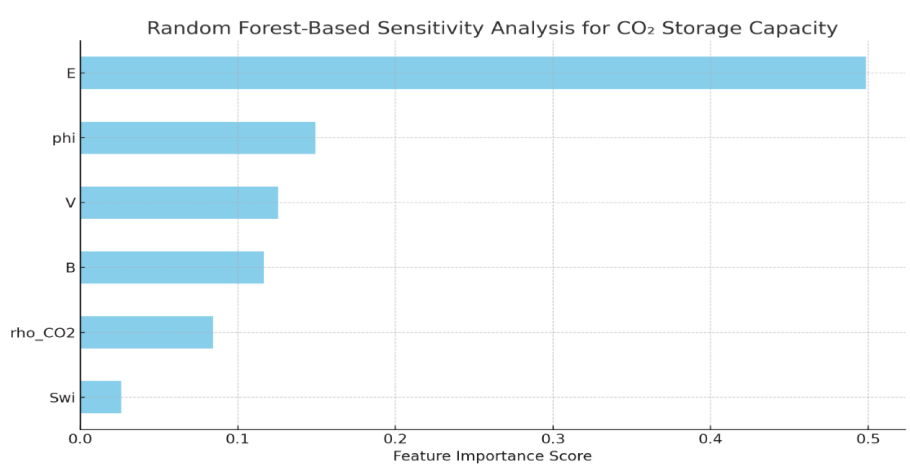

Figure 2—The Random Forest-Based Sensitivity Analysis for CO_2 Storage Capacity in the Mauddud reservoir

To perform the sensitivity analysis (both Monte Carlo and Random Forest-based), we needed to simulate a large number of plausible reservoir scenarios. This required defining realistic ranges for each of the parameters in the CO_2 storage capacity equation:

$$M\_CO_2 = V \times \phi \times (1 - Swi) \times B \times \rho CO_2 \times E$$

Here's how the ranges were selected and justified:

Gas Cap Volume (V)

- **Base Value:** 305,903,040 m^3 (from 248,000 acre-ft × 1233.48)
- **Range:** ±20% variability to account for volumetric uncertainty
- **Simulation Range:** 244.7 to 367.0 million m^3
- **Method:** np.random.uniform

Porosity (ϕ)

- **Base Value:** 0.31
- **Range:** 0.25 to 0.37
- **Rationale:** Based on observed variability in carbonate reservoirs
- **Method:** Uniform distribution around typical carbonate porosity values

Irreducible Water Saturation (Swi)

- **Base Value:** 0.11
- **Range:** 0.05 to 0.20
- **Rationale:** Covers variation in water saturation based on core and log analysis
- **Method:** Uniform distribution assuming conservative spread

Formation Volume Factor for CO_2 (B)

- **Base Value:** 1.5 m^3/m^3
- **Range:** 1.2 to 1.8
- **Rationale:** Reflects thermal and pressure variation under subsurface conditions
- **Method:** Uniform range used based on literature and PVT sensitivity

CO_2 Density (ρCO_2)

- **Base Value:** 600 kg/m^3
- **Range:** 500 to 700 kg/m^3
- **Rationale:** Accounts for variation in pressure-temperature behaviour of CO_2
- **Method:** Uniform distribution assuming supercritical CO_2 conditions

Storage Efficiency (E)

- **Base Value:** 0.5

- **Range:** 0.3 to 0.7
- **Rationale:** Typical range for saline aquifer or gas cap efficiency under structural and stratigraphic trapping
- **Method:** Uniform range consistent with IPCC/IEA guidelines

Monte Carlo simulations (Crystal Ball software)

To account for uncertainties associated with input parameters in the CO_2 storage capacity equation, Monte Carlo simulations were conducted using Crystal Ball software. This approach allows for the generation of probabilistic outcomes by assigning statistical distributions to each parameter—such as gas cap volume, porosity, irreducible water saturation, formation volume factor (FVF), CO_2 density, and storage efficiency. For each parameter, a logical range and distribution type (uniform or triangular) was defined based on literature values and engineering judgment. The simulation was run for 5,000 trials, resulting in a probability distribution of possible CO_2 storage outcomes. The resulting histogram illustrates the expected variation in storage capacity, with a P50 or mean of 37.57 MMt, and a P10 to P90 spread between 26.88 MMt and 49.47 MMt, capturing the model's inherent uncertainty, see Figure 3.

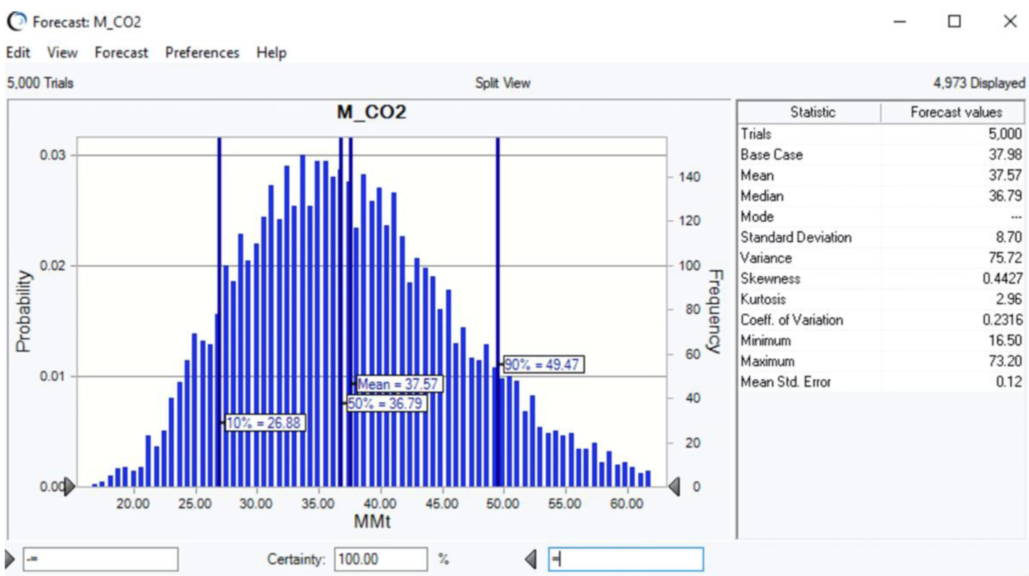

Figure 3—The probability distribution of possible CO_2 storage outcomes

Sensitivity analysis

Crystal Ball also provides a sensitivity diagram commonly referred to as a tornado chart that quantifies the relative influence of each parameter on the outcome (M_CO_2), see Figure 4. The software calculates the rank correlation coefficient between each input and the forecasted output, producing a visual representation of their impact. In this study, storage efficiency (E) was the most influential factor with a correlation coefficient of 0.73, followed by formation volume factor (FVF), porosity, gas cap volume, and CO_2 density. Irreducible water saturation (Swi) showed a mild inverse relationship (-0.11), as expected. This sensitivity analysis helps identify key drivers of storage uncertainty and guides future efforts in parameter refinement and data acquisition.

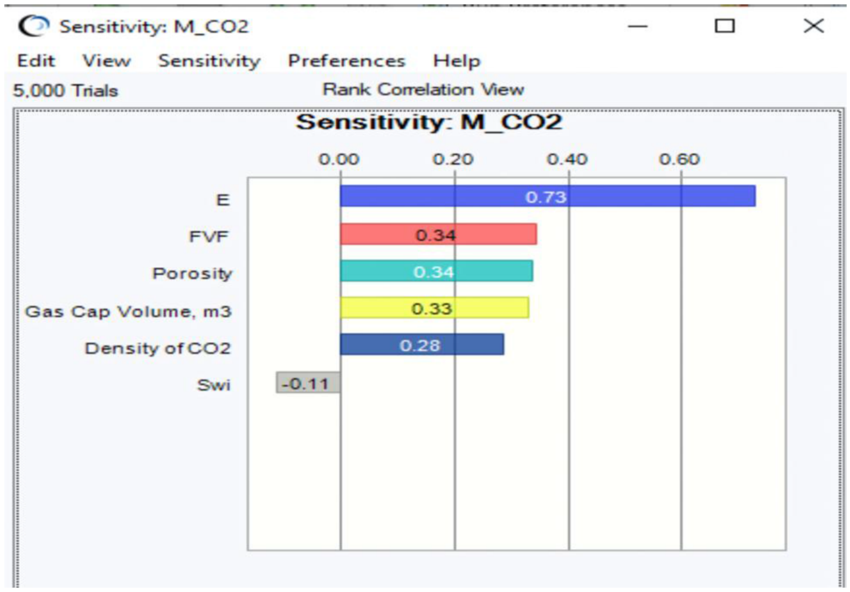

Figure 4—The sensitivity of each parameter on the outcome (M_CO_2).

Results

The volumetric analysis of the secondary gas cap in the Mauddud reservoir suggests a promising potential for CO_2 storage. Based on Monte Carlo simulations with 5,000 trials in Oracle Crystal Ball, the estimated mean CO_2 storage capacity is 37.57 million metric tonnes (MMt). This value was derived using realistic geological and reservoir parameters and reflects the central tendency of possible outcomes under defined uncertainty.

The simulation also provides probabilistic indicators:

- **P10 = 26.88 MMt** (conservative case)
- **P50 = 36.79 MMt** (most likely case)
- **P90 = 49.47 MMt** (optimistic case)

These values illustrate the forecasted range, and confidence bounds and help inform project planning under uncertainty. To assess which parameters have the greatest influence on CO_2 storage capacity, two sensitivity analysis methods were applied:

Crystal Ball Rank Correlation Sensitivity. Using rank correlation, the following parameters were ranked by influence:

- **Storage Efficiency (E):** 0.73
- **Formation Volume Factor (FVF):** 0.34
- **Porosity (ϕ):** 0.34
- **Gas Cap Volume (V):** 0.33
- **CO_2 Density (ρ):** 0.28
- **Irreducible Water Saturation (Swi):** –0.11

This analysis reveals that the storage efficiency (E) dominates, followed closely by FVF and porosity.

Machine Learning-Based Sensitivity Analysis

A Random Forest Regressor was trained on 1,000 synthetic samples using the same parameter ranges. The feature importance output closely matched the Crystal Ball results but with enhanced insight into non-linear relationships. The top influencers were:

- **Storage Efficiency (E):** most important
- **Porosity and FVF:** significant contributors
- **Swi:** minimal impact

The machine learning model confirmed and complemented the rank correlation findings and captured non-linear interdependencies, enhancing the robustness of the sensitivity assessment. In conclusion, the results support the technical feasibility of CO_2 storage in the Mauddud reservoir and highlight the critical role of reservoir efficiency, porosity, and fluid expansion properties. Cross-method validation strengthens confidence in both the estimate and its sensitivity to uncertainty.

Risk Evaluation

A comprehensive risk evaluation framework was developed to assess and manage potential hazards associated with CO_2 injection and storage in the Mauddud reservoir's secondary gas cap. The analysis was conducted using the Bowtie risk assessment approach (de Ruijter and Guldenmund, 2016), which systematically identifies threats, consequences, and corresponding mitigation measures. The key risk elements identified are as follows:

1. **Caprock Integrity**. The primary sealing unit may be vulnerable to mechanical failure due to pressure buildup during CO_2 injection. Additionally, geomechanical stress reactivation could compromise the sealing capacity. Continuous geomechanical modelling and injection pressure control are essential to maintain caprock effectiveness.
2. **Fault Reactivation**. The proximity of several structural faults to the planned injection zones raises concerns regarding fault slip and induced seismicity. Fault reactivation could potentially provide pathways for CO_2 leakage. This risk is mitigated by maintaining injection pressures below critical thresholds and by incorporating 3D seismic interpretation to map and avoid critical fault zones.
3. **Wellbore Integrity**. Numerous existing wells in the field were completed prior to the 1970s, and many may lack modern corrosion-resistant materials. These aging completions pose a risk of leakage through degraded cement or tubulars, especially under acidic CO_2-rich conditions. Risk mitigation includes wellbore diagnostics, zonal isolation techniques, and the use of corrosion-resistant alloys (CRA) in newly completed wells or workovers.
4. **Plume Migration and Containment**. If not closely monitored, the CO_2 plume may migrate toward structural flanks, potentially breaching containment structures. Risk is minimized by implementing real-time monitoring systems, including downhole pressure gauges and surface tiltmeter arrays, coupled with dynamic plume modelling to track migration pathways and ensure containment within the designated trap.

Mitigation Strategies

To address these risks, a suite of engineered and operational safeguards has been proposed:

- Zonal isolation through selective completions and cement squeezes.
- Real-time pressure surveillance with downhole gauges.

- Automated injection control protocols to limit pressure leakage.
- Use of corrosion-resistant completions and periodic well integrity testing.

This proactive risk assessment ensures that the CO_2 storage project remains technically viable, environmentally safe, and compliant with international carbon storage standards.

Conclusion

The assessment of the Mauddud reservoir for CO_2 storage in the Bahrain Oil Field demonstrates substantial technical potential and manageable risk, provided that appropriate geological, operational, and monitoring frameworks are in place. The volumetric and probabilistic analysis estimates a mean storage capacity of approximately 37.6 million metric tonnes (MMt), with the potential range defined by P10 and P90 values offering valuable insight into uncertainty margins.

Key risk elements including caprock integrity, fault reactivation**,** wellbore degradation, and plume migration were thoroughly evaluated using a Bowtie risk assessment approach. This framework facilitated a clear understanding of cause-consequence pathways and informed the development of robust mitigation measures, such as zonal isolation, injection pressure control, and continuous plume monitoring.

Additionally, the integration of Monte Carlo simulations and Machine Learning-based sensitivity analysis provided a dual perspective on parameter influence, highlighting storage efficiency, porosity, and formation volume factor as the most critical drivers of storage performance. This multi-tool analytical strategy enhances the decision-making process, ensuring confidence in both storage estimations and risk control strategies.

In conclusion, the Mauddud reservoir offers a viable and strategic target for CO_2 sequestration in support of Bahrain's decarbonization goals. With continuous surveillance, engineered safeguards, and targeted characterization of high-risk parameters, long-term secure storage of CO_2 can be achieved with minimal environmental or operational risk.

Nomenclature

CCS	**Carbon Capture and Storage**
FMI	**Formation microimager**
TVD	**True Vertical Depth**
API	**American Petroleum Institute**
OOIP	**Original Oil in Place**
GIP	**Gas in Place**
FVF	**Formation volume factor**

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